Chutes and Ladders: Institutional Setbacks on the Computer Science Community College Transfer Pathway

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Community colleges play a large role in educating students who are historically underrepresented in computer science (CS), including women, Latino men, and Black men, as well as post-traditional (older or working) students. In spite of this, there is a dearth of research on the institutional factors that influence whether or not community college students who are enrolled in CS classes and who express an interest in transferring and completing a bachelor's degree in the field persist. The overused "pipeline" metaphor, which indicates a supply-side lack, has been replaced by many with that of a "pathway." However, the "pathway" image suggests a general forward-moving trend that can be misleading. In this work, we draw from qualitative interviews with 14 CS students from groups traditionally underrepresented in the field who have studied introductory computer programming at a community college to investigate the following question: "What are the institutional barriers along a CS bachelor's degree track that includes community college?" Our findings indicate that there are three categories of institutional barriers along the transfer pathway: setbacks that hinder student progression forward, discontinuities in which students leave and re-enter the pathway, and departures in which students leave computer science and/or leave college altogether. We describe specific examples of each and introduce the idea of student movement as a game of "chutes and ladders," a convoluted trail where students can slide backwards or off the path (chutes), necessitating the implementation of targeted institutional supports that can boost student progress forward (ladders). We suggest institutional interventions that can help students facing each type of barrier to continue on course through community college and transfer to a four-year university.

CCS Concepts: • **Social and professional topics → Computer science education**;

Additional Key Words and Phrases: Community college, computer science, transfer

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1 INTRODUCTION

In recent years, nationwide attention has focused on the fact that many Science, Technology, Engineering, and Math (STEM) fields lack diversity, and computer science (CS) has had one of the biggest gaps; among CS undergraduates, 14.1% are female, 3.2% are African-American, and 6.8% are Latino [28, 29, 36]. While researchers have been motivated to study this disparity and identify ways

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to overcome it, far too often overlooked is the role that community colleges (CCs) play in postsecondary education. In preparation for those jobs that require a bachelor's degree, one route includes taking classes at a CC before transfer. With CCs serving a broad diversity of students—including large numbers of students traditionally underrepresented in CS—it makes sense to investigate factors that impede progress to transfer in CS.

2 BACKGROUND

Although community colleges in the United States were originally established to build a local workforce through job credentialing and continue to attract students with that goal, they are also being tapped by students as a less expensive option to complete their general education and lower division courses before transferring and completing a bachelor's degree at a baccalaureate college or university. As open access and lower-cost institutions, community colleges offer a more accessible gateway to students who cannot or choose not to enter directly into a four-year baccalaureate institution. CCs attract students with a range of academic capabilities, from those whose academic strength led them to graduate from high school early to those who lack the grades or preparation to apply to a 4-year university. Community colleges serve large numbers of students traditionally underrepresented in CS; 41% of all U.S. undergraduates—including 52% of Latino and 43% of Black undergraduates—are educated at CCs [30]. Close to half of students nationwide who receive a bachelor's degree first enrolled at a CC (https://nscresearchcenter.org/wp-content/uploads/SnapshotReport17-2YearContributions.pdf). Compared to students at 4-year baccalaureate institutions, CC students are older, more likely to be working full time, and more likely to have children [21].

Due to their wide range of backgrounds and goals, many students at CCs differ from the population of students interested in gaining entrance directly into 4-year baccalaureate institutions. In addition, the need to transfer from one institution to another as part of a bachelor's degree pathway to CS adds a step—and possibly even transfer shock [8]—to the pathway not required of those students who enter directly into the baccalaureate institution. Differing student populations and an extra admissions requirement step means that we cannot assume that findings on setbacks and opportunities for CS students at CCs are the same as findings for students directly admitted into CS at baccalaureate institutions, although findings from such research are a valuable starting point for further research that extends to CC populations. Similarly, research conducted on barriers and supports to students in STEM—particularly when conducted with students at CCs—can be used as a basis for research on CC CS students but cannot be assumed to be true for those students, since other STEM fields differ from CS in many critical aspects such as K-12 required exposure to the field (e.g., math is required but CS is not) and laboratory assignments (e.g., CS is lab-based; math is not).

To this point, most of the research done on student persistence in CS and/or STEM has focused on the contributing individual-level factors (for some exceptions, see work such as Rath, Rock, and Lafierre [32] and Chen, Mora, and Kemis [2]). In addition, much of the work uses statistical analysis of quantitative data to identify the factors that predict whether or not students persist rather than the perspectives or stories of the students. As a result, these studies have led the field to conclude that transfer pathways from community colleges into computer majors at universities are linear, decontextualized, and forward moving. In this journal, for example, are several works focused on issues in students'pathways into and through CS at the postsecondary level. Luse, Rursch, and Jacobson [19] used social cognitive career theory to investigate factors in student choice of an IT major, which give some explanation of individual factors that have a positive impact (e.g., interest and outcome expectations), but the study was done with university students, and it does not explore contextual factors for the decision. Pappas et al. [27] explore causal patterns of factors that influence 4-year university students' continued intention to study computer science, finding that it is critical to reduce such individual barriers as low student personal values and satisfaction with learning effectiveness to increase chances that they stay in the field. Barriers unique to CC students are not included in their model. McGill, Decker, and Settle [23] found that females who had participated in outreach computing activities were more likely to major in computing, but whether or not participants ever attended a CC is not mentioned.

In work reported outside of this journal, Denner, Werner, O' Connor, and Glassman [5] found that the factors that predicted intention to persist in CS at CCs were preparation and interactions with professors (for male students) and motivation, relational and behavioral factors such as peer support, expectations for success, and computer gaming (for female students). In one recent work focused on CS at CCs, students who completed transfer and a bachelor's degree in CS did less college "swirling" (lateral transfer) and lived in close proximity to a technology hub [14]. In another study, findings indicated that underrepresented CS students struggle to move efficiently through the CS chain of classes in preparation for transfer and are confused about transfer pathways [20]. For decades, successful baccalaurate attainment through a community college transfer pathway has been found to be in reach generally for academically advantaged students [18] who are fortunate and focused [14].

In contrast to previous work, this project investigates individual pathways from the perspective of the student to understand the confluence of institutional barriers that created setbacks along these pathways. From these retrospectives of what happened over time, we build a model based on larger themes that more closely reflects the lived experiences of CC CS students, which includes backwards, looping, and termination movement as well as a forward progression. As noted by Lee [17], investigation into real-world contexts allows an expansion of understanding how people learn beyond simply cognition to understanding what influences persistence and resilience. In addition, contextualizing and taking into account the intersectionality [16] of each student's experiences helps avoid the monolithic view by social category of much of the research done on underrepresentation. Only by understanding the life experiences of those who are following a CC CS pathway will we recognize what changes will most likely encourage a diversity of students to remain in the pipeline through the completion of a bachelor's degree. Although we recognize that students attend CCs for many reasons, this study was focused on transfer and completion of a bachelor's degree in CS, because that is the minimum education required for many CS jobs and careers. The goal of this work was to understand CS pathways from the point of view of the students to uncover and expose institutional factors that lead to setbacks. The results can be used to make recommendations to CCs, receiving 4-year institutions, and policy makers for specific changes that could move students forward towards transfer and a bachelor's degree in CS. Our research question for this study was as follows: What are the institutional barriers along a CS bachelor's degree track that includes community college?

3 RESEARCH METHODS

3.1 Participants and Procedures

As part of a larger study [20], qualitative interviews were conducted with 14 community college students traditionally underrepresented in CS (women, Latino men, and Black men) who had enrolled in introductory programming and who indicated on a survey that they intended to transfer and complete a bachelor's degree in CS.

To provide a retrospective on pathways through transfer, participants were identified from a sample that participated in an earlier study [5] and were interviewed 5 years after their enrollment in introductory programming. Forty participants who had reported 5 years earlier that they were "probably" or "definitely" intending to transfer and major in CS and who were from underrepresented groups were contacted via email and invited to participate in the study. The 14 participants who accepted are the focus of this article. These participants included three Asian-American females, a Black female, three mixed-race males (Black/Asian/White, Black/Latino/White, and Black/Native-American/White), one Latina, four Latinos, and two White women. This research was conducted in California; with more than 2.1 million students on 114 campuses, the California Community Colleges is the largest system of higher education in the United States (http://californiacommunitycolleges.cccco.edu/PolicyInAction/KeyFacts.aspx).

Since we were interested in investigating the students' perspectives on their pathways, we used a case study methodology. This work follows Yin's [35] definition of "an empirical inquiry that investigates a contemporary phenomenon (the 'case') in depth and within its real-world context, especially when the boundaries between phenomenon and context may not be clearly evident." (p. 16). Interviews were semi-structured, with prompts such as "What did you do after you left the community college [in which you took the survey]?," "Did you end up receiving a degree at the school?," "What were barriers to continuing in computer science?," and "What are your plans for any future education?" The interviewer remained open to stories related by the informant about barriers along their pathway, which allowed participants to share historical and descriptive information [35]. Interviews were conducted over the phone by the primary researcher—a White female with a background in CS—and audio recorded for ease of a word-for-word transcription. Interviews lasted from approximately 25 to 60 minutes. Participants received a small gift card by email as an incentive.

3.2 Data Analysis

Data analysis was done by open coding within each interview question [34]. The primary researcher is a White female with a MS in CS and a PhD in Learning Sciences; she has attended several CCs. As a first pass of data analysis, she coded several interview transcripts using Dedoose qualitative data software. Separately a Latino male researcher who has a master's degree in urban and regional planning and worked as staff at a CC coded the same interview transcriptions using a separate account. In both cases, themes about barriers and supports were allowed to emerge from the data in response to interviewer questions as well as additional topics that participants brought up. After this pass through the data, codes were compared between the two researchers and a common set of codes created after discussion and agreement between the researchers. The primary researcher then coded all transcripts using the updated codes. For purposes of this article, codes were extracted that were related to institutional barriers to progress along a CS pathway. These codes were then grouped by similarity into higher order themes and those themes divided into student outcomes—either setbacks but continuation on the pathway to a bachelor's degree in CS, departing from and returning to CS or to college, or leaving the pathway completely through leaving computer science or leaving college. Case studies illustrate the themes—an appropriate method for exploration and description of a phenomenon [24], with each "case" being a single participant from our study. Note that all names used are pseudonyms to preserve confidentiality. From the barriers found, corresponding supports are given as suggestions for targeted interventions that would help students move forward in a timely manner to transfer and completion of a bachelor's degree in CS.

4 FINDINGS

Findings from this study give insight into what the students view as barriers that slide them backwards or off the CS transfer and bachelor's degree completion pathway. Of the 14 students who participated in this study, only one had completed a bachelor's degree in CS five years after

Table 1. Barriers and Supports

Barrier	Examples	Suggested Supports
Pathway Setbacks	Incorrect academic advising	CS-focused counseling
	High GPA bar for transfer	Opportunities for grade forgiveness
	Refusal to accept transfer of classes	Flexibility (in criteria for incoming credits)
	Poor instruction	Support effective faculty
Pathway Discontinuities	Incompatibility of work and school	Accommodations for working students Flexibility (in class times and locations)
	Lack of alignment across CCs	Ease of college swirling Ease of re-entry
Pathway Departures	Lengthy math chains	Shortening/intenstifying the math chain
	Lack of college guidance	Readiness for college CS
	Lure of jobs	Career information
	Lure of other majors	Mesh with other majors

enrollment in introductory programming. Two others were on track to complete a CS bachelor's degree but were still in school. The rest of the participants were engaged in activities that deviated from the narrow pathway that was the focus of this study, from working on or completing certificates and degrees in fields related to CS to dropping out of college all together without receiving any certificate or degree.

The institutional setbacks experienced and described by participants in this study grouped into three categories (see Table 1). In some cases, students were delayed in their progress toward a CS degree by requirements to repeat classes or by misinformation—factors such as poor course descriptions, instructors not following the syllabi, or poor counseling—resulting in classes taken that did not further their progress ("Pathway Setbacks"). In other cases, students left school for periods of time and then returned to study CS, sometimes at a different CC ("Pathway Discontinuities"). Finally, some students changed to majors other than CS or had left college completely without receiving any certificates or degrees ("Pathway Departures"). We first describe each of these larger themes with examples gleaned from the data, then use four cases to illustrate some examples in context.

4.1 Pathway Setbacks

Participants in this study had experiences that pushed them back to an earlier stage along the CS pathway resulting in slower or aborted progress towards their bachelor's degree goal. Four students experienced these barriers, which included incorrect academic advising for transfer, high GPA requirements for transfer acceptance, non-transferability of courses between institutions, and poor instruction requiring a re-taking of classes. Despite experiencing one or more of these setbacks, three of the four participants had obtained or were in the process of obtaining a bachelor's degree, one in CS, one in Computer Engineering, and one in Technical Management; the fourth had departed from the CS bachelor's degree pathway after experiencing setbacks.

4.2 Pathway Discontinuities

Some of the participants in this study stepped out of college for a period of time and then returned to the same institution or swirled among multiple institutions. Three students had these discontinuities, two because of an incompatibility between work and school, and one because she re-entered college at a different CC.

4.3 Pathway Departures

For purposes of this study, a pathway departure was considered to be a departure from college before receiving a bachelor's degree and/or departure from CS (whether or not a bachelor's degree was completed). Students in this study experienced both setbacks and departures in some cases. Five years after enrolling in an introductory programming class with a stated intention to transfer and complete a bachelor's degree in CS, 10 participants had either left college and not yet returned—although all expressed an interest in "someday" returning for a bachelor's degree—or had left CS for another major. One student left CS after a few short weeks in the introductory class when she heard that CS had a calculus requirement and she had tested into remedial math, illustrating community colleges' uneasy balance between the roles of gate*way* and gate*keeper* [6]. Four other participants had obtained jobs that used their CS skills—one participant had completed an AA degree, two others certificates, and the fourth had no certificate or degree—and had been satisfied so far with their jobs and had not returned to college. Five students had changed majors to fields such as Economics/Finance, Library Information Science, Technical Management, Network Communication Management, and Information Technology Systems Security; of these, three had completed a bachelor's or master's degree, one had completed an associate's degree and was working on a bachelor's degree, and one had completed an associate's degree.

5 ILLUSTRATIVE CASES

The following cases provide more detail and context to the three categories described above. The stories participants tell about their pathways help to illustrate how they experienced the barriers and the impact these had on their original goal of pursuing a bachelor's degree in CS. These four cases were selected, because they describe a range of histories and narratives, as well as a range of outcomes.

5.1 Luis

Luis was a 28-year-old, U.S.-raised Latino who had completed an AA in CS and was working on a BS in CS at the time of this study. His pathway setback was the result of poor advising, but it was prevented from being a departure due to his passion for CS and his persistence in seeking ways to continue toward his goal.

Luis landed in CS through an early interest in computers, and he started his college career at a CC for financial reasons. Luis stated he had "always been fascinated with computers, even since I was 10 years old when we bought our first Windows 98 machine," and when Luis was 17 and in high school he built his first computer. Luis attended a community college after high school, because it was more "economical" and because it allowed him to "explore" different majors. He acknowledged that a 4-year baccalaureate institution would have had an advantage that students "graduate in three, four years because they' ll keep you on a straight path, versus a community college is kind of a very elusive path," but he was proud of his CC education, which he felt gave him more confidence; "just give me something to do, throw something at me and I' ll figure it out." While at the CC, Luis enrolled in introductory programming to learn the software side of computing and because "I didn't know what I wanted to do yet but I know I wanted to do something in computer science." He continued taking computer science classes, earning a certificate in computer programming and then an Associate's Degree in computer science.

Despite his drive to save money, Luis was forced to transfer to a private school to complete his bachelor's degree due to poor academic advising. His applications to public schools were all rejected because he was "missing an English class" for transfer, yet he could not take that class at the CC, because he had reached the maximum number of classes that they allowed at the school. Luis attributed this difficulty to the fact that the CC counselors "kind of messed up my academic plan, years ago; they had thought I only wanted to get an Associate's Degree, but I wanted to transfer out." Becoming desperate, he finally talked to an instructor who suggested that he apply for transfer to the private university, which he did successfully; the school allowed him to take the missing English class after transfer. Due to the high tuition cost of the private university, however, at the time of this study he was only taking two classes at that institution while completing additional classes online through public universities. He used this strategy to save money while fulfilling his general education classes (e.g., math and physics) but took his core CS classes at the private university; per Luis, public schools are "about a third of the price" of the private school.

In summary, Luis' pathway was made more difficult by poor advising. With CC pathways complicated by differing course requirements based on student goals, correct advising can be critical for student progression. The miscommunication between Luis and the advisor cost him the opportunity to transfer to a state-supported institution, which resulted in higher than necessary tuition. This turned out to be a setback rather than a discontinuity in the CS pathway due to Luis' workaround, which was to attend three different institutions at the same time to make his college education more cost effective.

5.2 Kylie

Kylie was a 28-year-old Asian-American woman working on an accelerated master's degree program in computer engineering, which would result in her earning both a BS and a MS in the field. Her setback was due to rigid institutional requirements and poor advising, but was prevented from being a departure by exposure to role models and being part of a larger CS community, which helped her to reject gender stereotypes about who does CS. She also cited supportive instructors and financial aid as playing a key role in helping her continue toward her goal.

When Kylie was younger, she was impressed by her brother's ability to build computers and she credits that with her later choice to enter the male-dominated field. Despite her early exposure, Kylie did not develop an interest in CS until later in life piqued by a conversation with a friend that left her wondering how software was made. Although she floundered at community college immediately after high school—her only postsecondary education choice—Kylie later enrolled in another CC to change jobs and pursue computer science, earning a certificate in C/C++ programming. Her pathway illustrates how institutional policies did not allow for the flexibility that she required to move forward efficiently and smoothly.

Kylie lacked motivation in high school, finishing "literally last" in her class and taking adult extension classes to finish what credits she needed to receive her diploma. She reported that "there was no way I was ever going to be able to get into a university out of high school. I didn't even bother to take my SATs." She enrolled in classes at a local CC after high school but continued to be unmotivated, saying that she "actually did end up failing one class because I never went to class." It was after this time that Kylie discovered an educational direction through an interest in CS. She attributed the catalyst for this interest to a conversation with a friend about programming when she suddenly wondered "how does all this software get made? Somebody has to program the games that I play." When she subsequently investigated CS, she realized that programming was similar to a logic class she had taken and liked. With her interest piqued, Kylie "actually started teaching myself how to program and realized I really liked it." She did this programming on her own time, "mainly just doing the basic stuff, like C, C++." She taught herself by reading books and through a chat room where she "started talking to all these people who were super passionate about programming. They were giving me little assignments or they would answer my questions.

I think that really helped." Kylie was not deterred from entering this male-dominated field because of role modelling by her brother when she was young. He was "really into computers," including building them, which impressed a young Kylie. Later, CS "didn't really seem like something that was a weird thing to know about because I had my brother. I never really thought it was a man versus a woman type of job or major. I guess I never really got exposed to the whole idea of 'Women don't really major in computer science or engineering.'"

To pursue her interest in CS, Kylie moved away from her hometown to an area with many computing jobs where she enrolled in a well-known CC that had a high transfer rate. Although it launched her on her CS pathway, this change of location and resulting enrollment in a different community college led to a setback on Kylie's pathway, since she wanted to transfer credits from her former CC to put her ahead in school. However, this meant that her grades from her classes—including for courses she had stopped attending but had not dropped—were also transferred. Kylie bemoaned that her lack of drive at an earlier time in her life meant that "those classes actually ended up hurting me the most... Those classes just stay on your record." Her later community college "didn't tell me I had to go retake that course at the school that I had failed it at in order to use the grade forgiveness; now that average is forever averaged into my GPA, so that kind of sucks." During her tenure at the community college, Kylie earned a certificate in C/C++ programming, because "it just so happened that I had already taken all the classes necessary for the certificate; I figured it would be a good thing to have."

Despite the problem of low grades in her early CC classes, at the time of the study Kylie had transferred to a state university and was pursuing an accelerated master's program in computer engineering where she was taking "my masters classes with my undergrad classes. It just lengthens my stay here for a year but then I get both degrees at the same time when I finish." When asked what might have stopped her from pursuing a bachelor's degree, Kylie responded that "probably for me the biggest concern was being able to pay for school. Honestly, if I didn't have financial aid there's no way I would have been able to do it. Even in working and going to school right now it's hard enough, but having to work enough to make enough money to pay for the entire tuition would be just really, really difficult. I probably wouldn't have been able to do it." When asked what encourages her to continue on her path to a master's degree, Kylie responded that she "just really enjoy[s] it. That's all, really, I can say is that I enjoy it. Otherwise, it wouldn't really make sense for me to keep doing this." She had found her instructors "really encouraging. I've never had a professor tell me, 'No, you shouldn't do this.'"

Kylie's pathway illustrates two pathway setbacks. First, because she made a lateral transfer ("swirl") between institutions, she had disadvantages in not being able to re-take classes for grade forgiveness. Second, her story hints at the importance of GPA in preparation for transfer.

5.3 Felipe

Felipe was a 40-year-old Latino who grew up in Panama and moved to the U.S. when he was 15 years old. He was working in a full-time programming job at the time of this study, although he had so far been unable to reach his goal of a BS in CS. His pathway setbacks, pathway discontinuities and pathway departure exemplifies how students can get sidetracked along their pathway by ineffective instructors and how close students can be to dropping out of college when there is any disturbance in their financial situation.

Felipe developed an early interest in programming. Not only did he become fascinated with computers and games when his parents brought a computer home, but—perhaps more unusual for the time—he reported that he "was also playing around with very simple code when I was a little kid. Nothing fancy. I was interested in code from an early age." Felipe attended a few semesters at a CC after high school but lost interest and left school and went to work. He landed at a nonprofit organization where he described part of his job as "small programming tasks for me to automate some of the daily tasks that I have to do." Friends that knew of his interest encouraged him to return to school and pursue CS, so Felipe went back to community college with the goal of earning an associate's degree, transferring to the local state university, and obtaining a bachelor's degree in CS. Felipe believed that starting back at a CC was "a great idea," finding "the amount of work and the amount of material and the amount of knowledge that you get from a junior college computer science curriculum, it's really, really good," while, at the same time, "a person will be saving a lot of money getting some of the units, some of the credits that they need to transfer to a four-year university." Felipe worked full-time and took his CC classes "evenings, weekends, and online." Felipe soaked up all the CS knowledge offered at the CC, reporting that he "took all the programming classes that I wanted to take up to the so-called advanced level that I could take at the college. It also left me wanting more. I learned so much that it triggered so many more questions and so many more interests." Along his pathway, Felipe earned certificates in Java programming and in Android programming. Meanwhile, he had not completed his general education requirements at the college, still needing "some of the general classes required to transfer . . . math classes and science lab, some physics classes."

Despite his enthusiasm for community college, Felipe added one caveat about CS instructors to our interview when asked if he had anything that might be interesting for the researchers to know. "Yeah . . . I did take a couple of classes where the instructor . . . was fairly old and wouldn't speak clearly and also would fall asleep in class. That person would have a TA to help them set up slide presentations and things like that. The TA would have to wake him up, or would have to explain why students were asking, if the students had questions. I did complain to the computer science department about this and head of the department said that there was nothing he could do about it due to the instructor's seniority in the school." This ended up delaying Felipe's progress, since his experience led him to drop the class twice, only completing it after enrolling a third time with "an instructor who knew his stuff, who was more awake, more aware and who would actually teach you."

Unfortunately, by the time of this study, Felipe had left the CS bachelor's degree pathway. He lost his job at the nonprofit, which he reported "put a stop to going back to school, because of job hunting. Eventually I got an internship related to programming and that basically took up all my time and haven't been able to find the time to go back." When asked why the loss of his job required him to quit school as well, Felipe responded that it was because of "financial reasons. When I lost the job, it took me about two or three months to actually find this internship. During that time, I decided not to go to school in order to save money in case of an emergency." His internship led to a full-time programming position, but his job prospects were limited by the lack of a degree. He reported that "my employer has a policy of offering full-time salary positions to people who have degrees, and because I don't have a degree at the moment I'm an hourly employee. Every year and a half or so they ask me, 'How's your degree going, because we'd like to make you a salaried employee?' " He thought about returning to school "all the time," but had not yet taken any concrete steps to re-enroll.

Felipe's story illustrates that leaving a college pathway can extend from a discontinuity to a departure. Until he discovered an interest and a direction for his college major goals, Felipe drifted in and out of community college. Once he decided upon CS, Felipe could move more purposefully towards his bachelor's degree goal, and he chose to attend a CC again to save money. However, he was limited in the CS classes he could take at the CC before transfer, and he was delayed in his pathway by an ineffectual instructor. Unfortunately, he finally ended up leaving the pathway due to financial hardships of a job change.

5.4 Brittany

Brittany was a 26-year-old White female student who had completed a BS in Technical Management at the time of this study. She had always been interested in computers, acting as IT help for friends, family, and work. Matriculating directly into a 4-year university was not financially feasible for Brittany, and she started her postsecondary schooling at a community college. After a rocky start in introductory programming, Brittany decided on a CS major. However, the fact that many of her CC classes would not transfer—a pathway setback—and she would have more required classes and a longer time at college to complete a BS in CS, Brittany switched to another major, resulting in a pathway departure from CS.

Brittany had "always been interested in computers. Since I got my first computer, I have always helped people figure out computers." Upon graduation from high school, Brittany attended a community college, since that was financially her "only option." She enrolled in introductory programming her first semester, explaining that she "had been working, doing IT related things and I knew I wanted to continue somewhere doing something with computers. [CS] was the closest thing that I could come up with that I was interested in" as a major. Although she received a passing grade in the class, Brittany did not feel that she had understood the material well enough. "It was not clear to me. I felt like it was way too much information given and not enough focus on the programming concepts." This caused her to feel discouraged about pursuing CS, so she "moved away from it for a while, and started thinking about other opportunities." Eventually, however, her interest in computers pulled her back into re-taking introductory programming when she "realized that was really what I wanted, even if it meant struggling with that part of it a little bit. I went back to it to see how it went," and found "it was a very different structured class the second time I took it," and she "felt much more confident with it and continued on with it from there." Brittany received several associate's degrees at the college, including computer science, social and behavioral sciences, and natural sciences, since earning the degrees did not require any classes in addition to what she was taking and she "wanted to have that flexibility if I happened to get a full-time job with just the associate's degree. I wanted that flexibility to not necessarily have to transfer if I didn't need to." After finishing at the community college, Brittany transferred to a for-profit university, where she "did about a year and a half of online classes there and then got my bachelor's" degree. She chose this university "mostly because of the flexibility with the online programs that they had. They were very robust online. They've been around for a while and doing it online allowed me to continue working [in technical support] which I needed to be able to do to continue going to school." Brittany described the important role that the university advisors played in helping her transfer successfully. "It was a little bit of an unusual situation and I was customizing a program to fit my needs so there was a lot of back and forth there. They were helpful figuring all that out." The program she chose was "technical management . . . I was particularly interested in digital and computer forensics, so they figured it out so that I could take classes in those areas and still have the degree in technical management." Brittany's interest lay in forensics and law enforcement, which a technical management degree allowed her to pursue.

When asked what would have been different if she had matriculated directly into a baccalaureate institution instead of choosing a CC and transfer, Brittany responded that "if I had started at a four year, I might have actually completed my bachelors in computer science, rather than doing an associate's in computer science and then going into technical management." As it was, she had considered a bachelor's degree in CS when she transferred, but "it would have required a lot more classes and not as many of my classes would have transferred. It would have been a longer path and it wouldn't have allowed me to take those forensic classes. Which is why I ultimately decided to do it the way that I did."

Brittany experienced delays in her CS pathway when she had to repeat introductory programming—in part due to the structure of the class her first time through—and when classes from her CC did not transfer to the university. Although she chose community college and transfer for cost reasons, Brittany ended up at a for-profit university because of their flexibility in allowing her to take classes online and to customize her major. She ended up not pursuing a CS major at least in part because some of her classes did not transfer.

6 DISCUSSION

Findings from this study show that students at CCs who intend to transfer and complete a bachelor's degree in CS run into barriers that can be described as affecting students in any or all of three ways: pathway setbacks that hinder or delay progress, pathway discontinuities when students do not continuously remain enrolled in college or when students "swirl" among institutions, and pathway departures when students leave college or change to other majors related or unrelated to CS. These barriers are similar to what others have found in CC students across fields of study [31, 32]. As outlined in the previous section, there are many examples of institutional barriers experienced by participants in this study that could be lessened with a concerted effort by postsecondary institutions to take such examples and make changes that would smooth and straighten the CS pathways that students traverse. The institutional barriers that have emerged from our data suggest concrete interventions that could boost student retention on the pathway for students who are aiming to transfer from community colleges to complete a bachelor's degree in CS (see 1).

6.1 Supports to Alleviate Pathway Setbacks

6.1.1 CS-Focused Counseling. As found by Rodriguez-Kiino [33], correct and easily available counseling and transfer information were important to students in this study and would have been a critical support to keeping students efficiently moving through college pathways. This is particularly true in a field such as CS, where prerequisite hierarchies and impacted programs can make forward progress difficult [20], and for first-generation college students who lack outside resources to help navigate a college path. CCs and receiving 4-year universities would serve underrepresented students by offering increased counseling and counselors knowledgeable about CS, specifically—perhaps even faculty. This supports findings by Packard et al. [26] of the importance of helpful advisors in women's successful transfer from a CC to a 4-year college in STEM.

6.1.2 Support Effective Faculty. Several students talked about the important role of faculty encouragement in helping them stay on or get back on the pathway. Others have highlighted the importance of "transfer champions" [7] and strong connections with effective, caring, and motivated teachers [10] as well as positive "insitutional agents" [9] as particularly important for low income and minority college students. These faculty need support and recognition for the key role they play. However, it is inexcusable to have instructors at the community college level that "fall asleep in class." Although none of the participants in this study talked about the impact of microagressions and low expectations, those have been documented by other studies of college students in non-traditional fields [11]. Policies should be changed or implemented that allow the identification and training or removal of ineffectual instructors.

6.2 Supports to Smooth Pathway Discontinuities

6.2.1 Flexibility and Accommodations for Working Students. CCs have a high proportion of post-traditional students—many of whom are underrepresented in CS—who must work to finance their schooling [21]. Because of this, both CCs and receiving 4-year institutions need to make accommodations for these students if they are committed to assisting them in continuing on a CS pathway. As found by Jesse [15], for-profit postsecondary institutions are serving many post-traditional students by offering the flexibility in counting credits and course locations and times that post-traditional students need. With for-profit institutions under fire for predatory and misleading admissions (e.g., https://www.wsj.com/articles/forprofit-college-probe-expands-1389647449?tesla=y), public institutions should consider better serving post-traditional student populations that can least afford to fall into for-profit traps of high tuitions and low graduation rates.

6.2.2 Opportunities for Grade Forgiveness, Ease of College Swirling, and Ease of Re-Entry. GPA transfer requirements form a strong barrier to CS students [13]. With CC students leaving and returning to college multiple times [22], creative ways for allowing students both to change community colleges and to improve their GPA in preparation for transfer would smooth the pathway to a bachelor's degree in CS. In addition, accommodating student re-entry into college after time away would encourage post-traditional and underrepresented students to continue their education. In support of Jaggars et al. [14], college "swirling" was a setback on the pathway for students in this study. Some of these pathway issues can be addressed by a system-wide set of course-to-course equivalencies and a common course numbering system, as described by the Educational Commission of the States (https://www.ecs.org/transfer-and-articulation-policies-db/), examples of which have been implemented in Florida (https://flscns.fldoe.org/) and Texas (https://www.tccns.org/). However, some caution is warranted in these schemes, as similarly named and/or numbered courses can be taught quite differently, particularly at different types of insitutions. Regional partnerships between institutions may also implement course equivalencies, articulation agreements, reverse transfer, and so on, in planned ways that support student transfer and success between the partnering institutions.

6.3 Supports to Discourage Pathway Departures

6.3.1 Shortening/Intensifying Math Chains. Institutions interested in keeping a diversity of students on the CS pathway will want to address the math requirements. K-12 programs designed to entice underrepresented students to CS can leave CC students high and dry if other required classes such as math are not emphasized to interested students. As found by Rodriguez-Kiino [33], anticipated extended times to degree attainment in CS lead to pathway departures for students in our study. Many students enter CCs testing at developmental math levels [1]; offering intensive math catch-up opportunities could encourage students to pursue or continue in CS even if they need to catch up in math. In addition, information for students about majors related or unrelated to CS that require less math might keep students in a computing pathway, even if they decide to leave CS.

6.3.2 Readiness for College CS. First-generation college students need information and tools to help them navigate a college pathway, since informal home and community resources available to more privileged students are not available. Efforts are underway to solve the counseling problem in high school with state-wide CSforAll and CS Ed Week efforts, and NCWIT (ncwit.org) has a "Counselors for Computing" program. While some colleges offer bridge programs to ready students for college [3], students in computer science may need more targeted help to navigate the tight prerequisite chains required by the major and for transfer. In addition, information about careers available with a CS degree would be helpful not only to first-generation students, but to all students, since CS and related fields are not required learning in K-12 schooling.

6.3.3 Career Information and Mesh with Other Majors. The world of CS majors, degrees, and careers is complex. Two- and 4-year colleges offer career AA/AS and certificate programs, CS,

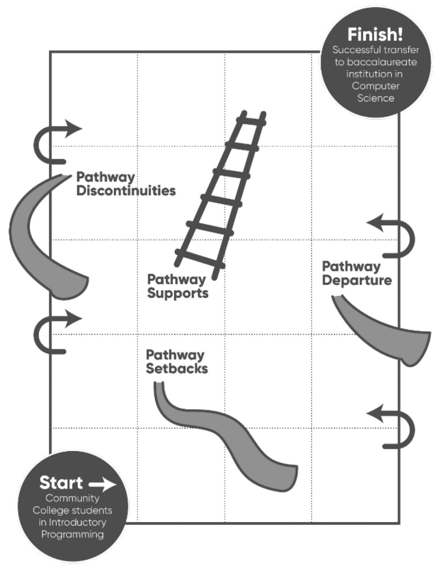

Fig. 1. Chutes and ladders.

CE, IS, IT, CS+X and BA/BS programs. Not only are there computing majors related to CS, but programming—a basic skill of CS—is useful or even required for other college majors. Students, particularly first-generation college students, may have a difficult time aligning a major with a career, as well as knowing what careers are possible with a CS major [20], and outreach to students and their families is important [25]. Helping students understand the career outcomes for CS graduates and graduates of other majors where programming is required could foster more direct pathways through CCs, transfer, and to bachelor's degrees. As suggested by Herrera and Hurtado [12] and Jackson [13], exposure to CS jobs encouraged students in their pursuit of CS. However, in the cases reported on here, it can also entice students off a bachelor's degree pipeline to enter directly into the workforce.

7 CONCLUSION: A NEW METAPHOR

In her work on nontraditional pathways, Jesse [15] argues that a "pipeline" is an inappropriate—and perhaps even harmful—metaphor for illustrating the opportunities and constraints that many students experience on their road into CS education and the workforce. This work supports her argument, and—based on the data we report on here—we propose a metaphor that better fits the experiences of these students—a metaphor we call "chutes and ladders" based on the popular board game where players experience both setbacks ("chutes") and shortcuts ("ladders") that influence pathways and time to the final, winning square (see Figure 1).

In the game of "chutes and ladders," players begin at square one on the game board and spin a spinner to move forward the number of spaces designated by the spinner along a road towards the goal of being the first to reach the end square on the board. Along the route, however, certain squares are marked as the beginning of a "chute" or the beginning of a "ladder." Players who land on a "chute" are forced to move backwards to earlier game squares; the number of squares back varies according to the length of the chute. Players who land on a "ladder" during game play are allowed to skip forward to later game squares. Analogous to these "chutes" and "ladders" are the constraints and opportunities that jump students forward or slide students backward along their pathway to completion of a computer science bachelor's degree. Unlike the game, some "chutes" along CS student pathways slide students off the road altogether. Crul [4] used a similar metaphor to describe the school trajectories of youth in Europe, where he found that the pathways to and away from success vary greatly across country. Like ours, his study shows the importance of tracking these trajectories and understanding the range and key points at which students veer off course.

Our findings support a metaphor such as that of "chutes and ladders" that indicate CC CS students' movements along the pathway to a bachelor's degree follows many directions: forwards, backwards, and even laterally. The setbacks that students experienced included both delays on the pathway and barriers that slid them off the pathway altogether. Unfortunately, these challenges can be missed when research does not distinguish between different types of STEM fields, or focuses on students in CS at 4-year universities. This study took a unique approach by investigating CC students' lived experiences in regard to an earlier intention to transfer and pursue a CS bachelor's degree. The results highlight both the institutional challenges they experienced, as well as the factors that helped some of them continue toward their goal.

8 LIMITATIONS

As is common with qualitative work, our goal was not generalizability to the community college student population as a whole, but rather to give context and to offer participant perspectives supported by quotes in order for the reader to judge similarity to other settings and other students. Since this study was conducted in California, particular caution is warranted when attempting to draw conclusions about community colleges in other states, which may be guided by different policies and regulations and which may serve a very different population of students; there is a wide range of community college systems in the U.S. (https://www.leg.state.nv.us/interim/77th2013/Committee/Studies/CommColleges/Other/28-January-2014/AgendaItemVI,NationalCenterforHigherEducationMcGuinness.pdf).

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